1 Quick refresher on the current in electrodynamics

1.1 Maxwell's equations

Given Maxwell's equations,

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \frac{1}{\varepsilon_0}\, \rho(\boldsymbol{r}, t) \tag{1}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0 \tag{2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial}{\partial t} \boldsymbol{B}(\boldsymbol{r}, t) \tag{3}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \boldsymbol{E}(\boldsymbol{r}, t) + \frac{1}{\varepsilon_0 c^2}\, \boldsymbol{j}(\boldsymbol{r}, t) \tag{4}$$

1.2 Continuity equation

From $\varepsilon_0\, \partial_t$ (*eq.* (1)), using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\boldsymbol{r}, t) = \varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big), \tag{5}$$

and from $\varepsilon_0 c^2\, \boldsymbol{\nabla} \cdot$ (*eq.* (4)),

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) &= \boldsymbol{\nabla} \cdot \big(\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) \big) - \varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big) \\ &= -\varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big) \qquad (\mathrm{div}(\mathrm{rot}\, \boldsymbol{F}) = 0). \end{aligned} \tag{6}$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\boldsymbol{r}, t) + \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) = 0. \tag{7}$$

1.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\boldsymbol{r}, t) = \sum_\alpha q_\alpha\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) \tag{8}$$

$$\boldsymbol{j}(\boldsymbol{r}, t) = \sum_\alpha q_\alpha\, \boldsymbol{v}_\alpha(t)\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) \tag{9}$$

Proof. Note that

$$\begin{aligned} \partial_t\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) &= \partial_t \Big(\delta\big(r_x - r_x^\alpha(t) \big) \delta\big(r_y - r_y^\alpha(t) \big) \delta\big(r_z - r_z^\alpha(t) \big) \Big) \\ &= -\sum_{i=x,y,z} \dot{r}_i^\alpha(t)\, \partial_{r_i} \delta\big(r_i - r_i^\alpha(t) \big) = -\dot{\boldsymbol{r}}_\alpha(t) \cdot \boldsymbol{\nabla}\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big), \end{aligned} \tag{10}$$

where we used the chain rule in the second line and

$$\boldsymbol{\nabla}\cdot\Big(\boldsymbol{v}_\alpha(t)\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)=\sum_{i=x,y,z}\partial_{r_i}\Big(v_i^\alpha(t)\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)$$
$$=\sum_{i=x,y,z}v_i^\alpha(t)\,\partial_{r_i}\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)=\boldsymbol{v}_\alpha(t)\cdot\boldsymbol{\nabla}\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big). \qquad (11)$$

Therefore

$$\frac{\partial}{\partial t}\rho(\boldsymbol{r},t)=-\sum_\alpha q_\alpha\,\dot{\boldsymbol{r}}_\alpha(t)\cdot\Big(\boldsymbol{\nabla}\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)=-\boldsymbol{\nabla}\cdot\boldsymbol{j}(\boldsymbol{r},t), \qquad (12)$$

which is (7). □

Note that these solutions are not unique (one equation for four components).

Intermediate result: the current can be defined by "charge times velocity".